① Aspect 13 (Most Central)

Use an understanding of the context to guide or expedite the mathematical solving process

oil-spill-qs1 is clearly embedded in a **real-world environmental context (an oil spill)**. Students are required to:

- Understand the realistic background described in the problem (e.g. pollution spread, affected area, changes over time)
- Judge which quantities and relationships are meaningful in real-world terms
- Apply **context-appropriate assumptions and levels of accuracy** in their calculations

Mathematical solving is therefore not carried out in isolation, but is strongly constrained and guided by the real-world situation. As a result, **contextual understanding is the primary driver of the solution process**.

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② Aspect 12

Use a multi-step procedure leading to a mathematical solution or conclusion

The question typically requires students to:

- First calculate one or more intermediate quantities (such as area, rate of change, or extent of spread)
- Then use these results to complete further calculations or reach a conclusion

The solution process is therefore **structured, sequential, and multi-step**, requiring careful organisation and execution.

③ Aspect 7

Manipulate numbers, graphical and statistical data and information

Students must:

- Process and operate on numerical information provided in the problem
- Correctly manage quantitative relationships such as growth, change, proportions, or unit conversions

Numerical and data manipulation skills form the **operational foundation** of the multi-step solution.

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④ Aspect 10

Make sense of, relate and use a variety of representations when interacting with a problem

oil-spill-qs1 often involves:

- A combination of written descriptions and numerical information
- Implicit geometric or spatial interpretations (such as regions, areas, or spreading patterns)

Students must connect these different representations and understand how they collectively describe the same real-world phenomenon.

⑤ Aspect 8

Articulate a solution, showing and/or summarising intermediate mathematical results

In a complete response, students are generally expected to:

- Show key intermediate calculation steps
- Clearly present the final conclusion and explain its real-world meaning (e.g. the size of the polluted area)

Successful performance therefore requires not only correct calculations, but also **clear communication of reasoning and results**.

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⑥ Aspect 1

Perform a simple calculation

Although the overall context and structure of the problem are complex, the individual calculations involved are typically:

- Basic arithmetic operations or the application of familiar formulas

As a result, this aspect plays a supporting rather than central role.

① Aspect 13（最核心）

Use an understanding of the context to guide or expedite the mathematical solving process

📌 *oil-spill-qs1* 明显嵌入在一个 **环境与现实问题情境（石油泄漏）** 中。学生需要：

- 理解题目所描述的现实背景（如污染扩散、受影响区域、时间变化等）
- 判断哪些数量和关系在现实中是有意义的
- 在计算时采用 **符合情境的精度和合理假设**

👉 数学求解并非脱离现实进行，而是受到情境强烈约束，因此 **情境理解是解题的首要驱动力。**

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② Aspect 12

Use a multi-step procedure leading to a mathematical solution or conclusion

📌 题目通常要求学生：

- 先计算一个中间量（如面积、变化量、扩散范围）
- 再基于该结果完成进一步计算或判断

👉 解题过程具有明显的 **步骤性和层次性**，需要学生正确组织并执行多步计算。

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③ Aspect 7

Manipulate numbers, graphical and statistical data and information

📌 学生需要：

- 对题目中给出的数值信息进行处理
- 正确操作涉及的数量关系（如增长、变化、比例或单位转换）

👉 数值与数据操作是完成多步骤求解的基础工具。

④ Aspect 10

Make sense of, relate and use a variety of representations when interacting with a problem

📌 *oil-spill-qs1* 往往涉及：

- 文字描述与数值信息的结合
- 可能隐含几何或空间意义（如区域、范围、扩散形态）

👉 学生需要在 **不同表征之间建立联系**，理解这些表征如何共同描述同一现实问题。

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⑤ Aspect 8

Articulate a solution, showing and/or summarising intermediate mathematical results

📌 在完整作答中，学生通常需要：

- 展示关键中间计算步骤
- 清楚表达最终结论，并说明其现实含义（如污染范围大小）

👉 解题不仅要求“算对”，还要求“说清楚”。

⑥ Aspect 1

Perform a simple calculation

📌 虽然题目情境复杂，但具体计算步骤本身通常是：

- 基本算术或熟悉的公式运用

👉 单个计算并不复杂，因此该方面的重要性相对较低。